Dhafer Malouche, Ph.D.,

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Google scholar page

Employment History

From April 2021 — **Pan-African Scientific Research Council**, Fellow member.

From October 2020 — **Covidradar24.org** and **Rosettahub.com**, Data analyst master contributor in a real time COVID-19 tracking data science platform.

05/2020 – 01/2021 — **World Health Organization, Tunisia** Senior data analyst consultant, Measuring the impact of the COVID 19 pandemic on psychiatric needs of the general population in Tunisia, data collected from a hot line accessible throughout the country, including those without access to Internet.

WARC, Africa, Sierra Leon Data analyst consultant, Setting up an online daily data monitoring platform for a survey implemented in Sierra Lean.

2003 – · · · · — **University of Carthage,** Ecole Supérieure de la Statistique et de l'Analyse de l'Information, Tunis, Tunisia

- **Positions:** Professor (from 2018), Associate Professor (2011-2018), Assistant Professor (2003-2011).
- **Teaching:** Data Analysis: Principal Component Analysis, Correspondence Analysis, and Multiple Correspondence Analysis, Theory and Practice with `R`, Mathematical Statistics: Statistical Inference, Hypothesis Testing, Regression Analysis, Theory and Practice with `R`, Data Mining and Practice with `R`, Bayesian Statistics with `OpenBugs/Winbugs/RStan/Jags`, Time Series: ARMA and SARIMA Processes, Theory and Practice with `R/Python`, Big Data: Large Data with `R/Python`, `SQL`, `Spark`, `H2O`, Advanced `R/Python`: Data Management, Data Visualization, Dashboards, Shiny Apps, Heroku Apps, Bokeh.
- **Research:**
  - Supervising 5 Ph.D.: Detection and classification of swallowing sound, Sensory Analysis, Genetics, Zoonotic cutaneous leishmaniasis incidence, Cardiovascular risk factor.
  - Papers and collaborations: I had published and coauthored more than 40 scholar papers.
  - Supervising Master and Engineering Thesis on several topics related to applied statistics and data science.
- **Administration:** Director of the Department of Statistics (2004-2007)

2014 – 2019 — **Yale University,** Whitney and Betty MacMillan Center for International and Area Studies and the Department of Statistics and Data Science, New Haven, USA.

- **Positions:** Visiting Associate Professor (2014), Consulting on several projects with Yale Scholars (2015-2018), Associate Research Scholar (2019).
- **Missions:** Democratic Transition in Tunisia, Governance and Local Development: Implementing two face-o-face Surveys in Tunisia, G-econ team on building local GDP data,Teaching Time Series with `R/Python Course`.

Employment History (continued)

2016 – 2017 **University of Michigan**, Center for Political Studies of the Institute for Social Research, Ann Arbor, USA.

- **Position:** Visiting Fulbright Scholar.
- **Role:** Research: Working on Data Quality, Survey Methodology, Interviewer Effect, Teaching: 4 Lectures on Applied graphical Models, 1 Lecture on Data visualization with R, 1 Lecture on Sensory Analysis.

2011 (May to July) **Stanford University**, Department of Statistics, Palo alto, USA.

- **Position:** Visiting Fulbright Scholar.
- **Role:** Research on Graphical Models, Faithfulness assumption, and Covariance graphs.

2002 – 2003 **York University**, Department of Statistics, Toronto, Canada.

- **Position:** Visiting Assistant Professor,
- **Teaching:** Applied Regression Models with SAS, Introduction to the theory of probability, Introduction to Statistics with Minitab,
- **Research:** Monte Carlo Methods and Bayesian Estimation of the Graphical Models.

1998 – 2002 **University of Sousse** , Institut Préparatoire aux Ecoles d'Ingénieurs, Sousse, Tunis.

- **Teaching:** Analysis, Calculus, Algebra
- **Research:** Natural Exponential Families, Pick functions, Markov Chains
- **Administration:** Director of the Department of Mathematics

Education

Sept 2009 **Habilitation *(Tenure)*, Université de Tunis El Manar** Ecole National d'Ingénieurs de Tunis, Tunisia.
Thesis title: *Problèmes autour de la probabilité et de la statistique: Méthodes et Applications.*
Dissertation: `https://malouche.github.io/myCV/reports.html`

October 1997 **Doctorate, *(Ph.D.)*, Paul Sabatier University,** Toulouse, France.
Thesis title: *Classification des familles exponentielles associées à des fonctions Pick*.
Dissertation: `https://malouche.github.io/myCV/reports.html`

1993–1994 **Master's Degree (D.E.A), Paul Sabatier University,** Applied Mathematics.

1989 – 1993 **Bachelor (Maîtrise)**, Ecole Normale Supérieure de Bizerte, Tunisia.

Skills

Languages Strong reading, writing and speaking competencies for English, French, and Arabic.

Coding Python, R, TABLEAU, SPARK, H2O, Shiny, LaTeX, ...

Misc. Data Mining, Machine Learning, Big Data, Academic research, Teaching, training, consultation, LaTeX typesetting and publishing.

Research Publications (Most recent papers)

Research interests: Public health, COVID-19, Research and development, Well being, Survey methodology, Consumer preferences, Genetics.

1. Lahmandi-Ayed, R., & Malouche, D. (2020). More investment in research and development for better education in the future? *Available at SSRN 3739632*.
2. Malouche, D., & Ben Romdhane, H. (2020). Estimating excess mortality associated with covid-19 pandemic: A 151 cross-countries study. *Estimating Excess Mortality Associated with COVID-19 Pandemic: A*, *151*.
3. Saidi, O., Malouche, D., Saksena, P., Arfaoui, L., Talmoudi, K., Hchaichi, A., … Ben Alaya, N. (2020). Impact of contact tracing, respect of isolation and lockdown in reducing the number of cases infected with covid-19: Case study: Tunisia's response from march 22 to 04 may 2020. *International Journal of Infectious Diseases*.
4. Kongbonga, G. Y. M., Hassine, K. B., Ghalila, H., Malouche, D. et al. (2019). Front-face fluorescence using uv-led coupled to usb spectrometer to discriminate between virgin olive oil from two cultivars. *Food and Nutrition Sciences*, *10*(02), 119.
5. Saidi, O., Hajjem, S., Zoghlami, N., Aounallah-Skhiri, H., Mansour, N. B., Hsairi, M., … O'Flaherty, M. et al. (2019). Premature mortality attributable to smoking among tunisian men in 2009. *Tobacco induced diseases*, *17*.
6. Saidi, O., O'Flaherty, M., Zoghlami, N., Malouche, D., Capewell, S., Critchley, J. A., … Guzman Castillo, M. (2019). Comparing strategies to prevent stroke and ischemic heart disease in the tunisian population: Markov modeling approach using a comprehensive sensitivity analysis algorithm. *Computational and mathematical methods in medicine*, *2019*.
7. Saidi, O., Zoghlami, N., Bennett, K. E., Mosquera, P. A., Malouche, D., Capewell, S., … O'Flaherty, M. (2019). Explaining income-related inequalities in cardiovascular risk factors in tunisian adults during the last decade: Comparison of sensitivity analysis of logistic regression and wagstaff decomposition analysis. *International journal for equity in health*, *18*(1), 1–11.
8. Salem, S., Malouche, D., & Romdhane, H. B. (2019). Tunisian population quality of life: A general analysis using sf-36. *Eastern Mediterranean Health Journal*, *25*(9).
9. Benstead, L. J., Kao, K., Landry, P. F., Lust, E. M., & Malouche, D. (2017). Using tablet computers to implement surveys in challenging environments. *Survey Practice*, *10*(2), 1–9.
10. Khouloud, T., Hedia, B., Nissaf, B.-A., Marc, S., Dhafer, M., & Kouni, C. M. (2017). Comparative performance analysis for generalized additive and generalized linear modeling in epidemiology. *International Journal of Advanced Computer Science and Applications*, *8*(12). doi:10.14569/IJACSA.2017.081255
11. Rebhi, I., & Malouche, D. (2017a). An approach for external preference mapping improvement by denoising consumer rating data. *INTERNATIONAL JOURNAL OF ADVANCED COMPUTER SCIENCE AND APPLICATIONS*, *8*(12), 500–508.
12. Rebhi, I., & Malouche, D. (2017c). Decision making about products development through consumer preferences modeling based on descriptive characteristics of products, 423–430.
13. Talmoudi, K., Bellali, H., Ben-Alaya, N., Saez, M., Malouche, D., & Chahed, M. K. (2017). Modeling zoonotic cutaneous leishmaniasis incidence in central tunisia from 2009-2015: Forecasting models using climate variables as predictors. *PLoS neglected tropical diseases*, *11*(8), e0005844.
14. Triki, H. Z., Laabir, M., Lafabrie, C., Malouche, D., Bancon-Montigny, C., Gonzalez, C., … Daly-Yahia, O. K. (2017). Do the levels of industrial pollutants influence the distribution and abundance of dinoflagellate cysts in the recently-deposited sediment of a mediterranean coastal ecosystem? *Science of the Total Environment*, *595*, 380–392.
15. Aouinti, S., Malouche, D., Giudicelli, V., Kossida, S., & Lefranc, M.-P. (2016). Imgt/highv-quest statistical significance of imgt clonotype (aa) diversity per gene for standardized comparisons of next generation sequencing immunoprofiles of immunoglobulins and t cell receptors (vol 10, e0142353, 2015). *PLOS ONE*, *11*(1).